
Software Requirements Specification

**For
DentaAI**

AI-Powered Dental Disease Detection and Doctor Consultation System

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1. Introduction

1.1 Purpose

The purpose of DentaAI: AI-Based Dental Disease Detection & Doctor Consultation System is to develop a web-based platform that uses artificial intelligence and deep learning to provide preliminary detection of dental diseases from uploaded intraoral images. The system uses a CNN-based transfer learning model to classify dental conditions and provides prediction probabilities and confidence scores to support early dental assessment.

The system allows patients to upload dental images, receive AI-based predictions, view previous assessments, and generate PDF reports. Patients can also browse verified dentists, select a suitable doctor, book consultations, make secure online payments, and receive consultation updates.

Doctors can access assigned consultation requests, review the patient's uploaded dental images and AI predictions, verify the results, provide diagnosis and treatment recommendations, and communicate with patients. Administrators can manage users and doctors, monitor consultations, and oversee overall system activities.

The system also supports Grad-CAM-based visualization to provide an interpretable view of the image regions that contributed to the model prediction. The system is intended as an academic research and preliminary screening prototype and does not replace professional dental examination or provide a final medical diagnosis without doctor verification.

1.2 Intended Audience

The intended audience includes:

- Doctors and dental specialists – To review AI results, verify diagnoses, and provide treatment recommendations.
- Administrators – To manage users, doctors, and system activities.
- Patients – To upload dental images, view AI predictions, and consult doctors.
- Developers and Machine Learning Engineers – To maintain and enhance the system. Software developers
- Researchers and students – To study AI-based dental disease detection and consultation systems.

1.3 Scope

The AI-Powered Dental Disease Detection and Doctor Consultation System is a web-based system developed to provide preliminary dental disease detection from uploaded oral images and facilitate consultation with dental professionals. The scope of the system includes:

- Collecting and preprocessing dental image data.
- Training a deep learning model for dental disease classification.
- Processing uploaded dental images and predicting supported dental conditions.
- Displaying AI-based predictions and confidence scores.
- Generating and managing dental scan reports.
- Allowing patients to select doctors and book consultations.
- Providing online consultation payment through Razorpay.
- Allowing doctors to review AI predictions and provide professional diagnosis and treatment recommendations.
- Managing patient, doctor, and admin activities through role-based access.
- Sending email and SMS notifications for registration, consultation, payment, and doctor approval.

The system is limited to preliminary AI-based dental assessment and consultation support and does not replace professional dental diagnosis or physical examination.

1.4 References

- **Hafeez et al. [1] (2022)** proposed Dental-Net, a CNN-based model for dental cavity detection from oral images. It achieved 91.09% validation accuracy, demonstrating the effectiveness of deep learning for automated dental screening.
- **Rathod et al. [2] (2025)** developed a CNN-based model for detecting common oral diseases such as plaque, calculus, gingivitis, and caries. The model achieved 81.11% accuracy, showing performance comparable to dentists.
- **Kühnisch et al. [3] (2021)** used MobileNetV2 for detecting dental caries from intraoral photographs. The study achieved over 90% agreement under ideal conditions and demonstrated the potential of transfer learning for dental diagnosis.
- **Felsch et al. [4] (2023)** proposed a Vision Transformer-based model for detecting and localizing dental caries and MIH. The model achieved an accuracy of 97.8%, demonstrating effective localization of pathological regions.

- **Adnan et al. [5] (2024)** developed a smartphone-based dental decay detection system using YOLOv5s. The system achieved 90.7% precision, 85.6% sensitivity, and 88.0% F1-score, showing the potential of AI-based dental screening applications.

2. Overall Description

2.1 Product Perspective

The system is a full-stack web application using React with Vite for the frontend and Node.js with Express for the primary transactional backend. MongoDB is used for structured data storage.

TensorFlow and Keras are used for deep learning. A deep learning model (e.g., CNN/MobileNetV2/ResNet50) is used as the primary image classification model. OpenCV, Pillow, and NumPy are used for image processing. Grad-CAM is used for explainability, and PDF generation libraries are used to generate PDF reports.

The frontend communicates with the Node.js backend through REST APIs. The Node backend handles authentication, patient records, consultations, real-time messaging, and database operations. An independent Python Flask backend handles MRI/X-ray processing, model inference, and Grad-CAM explainability generation.

2.2 Product Functions

The system shall provide:

- User authentication and role-based access.
- Doctor management by Admin.
- Patient registration and profile management.
- Dental image (intraoral scan/X-ray) upload and validation.
- Image preprocessing.
- Deep learning-based dental disease classification.
- Class probability calculation.
- Confidence and uncertainty analysis.
- Grad-CAM visualization.
- Patient AI report generation and history.
- Consultation booking and secure simulated checkout.
- Doctor diagnostic workspace and clinical override.

- Real-time chat between patient and doctor.
- PDF report generation and download.
- Audit logging.
- Demo/sample data accounts.
- Error handling and security.

2.3 User Classes and Characteristics

Admin: The Admin manages the overall system.

Admin can:

- Add and manage Doctors.
- Activate or deactivate Doctors.
- View registered Patients.
- View overall system consultations and assessments.
- Monitor system statistics and audit logs.
- Admin does not directly consult patients or register patients.

Doctor: The Doctor performs clinical verification and consultation activities.

Doctor can:

- Log in securely.
- Accept or decline incoming consultation requests.
- View the patient's uploaded images and primary complaints.
- View AI predictions, confidence scores, and Grad-CAM visualizations.
- Override or confirm AI results by submitting a Final Diagnosis and Treatment Plan.
- Communicate with patients seamlessly via real-time chat.
- View patient assessment history for accepted cases.
- Generate and download PDF reports.

Patient: The Patient seeks dental advice and screening.

Patient can:

- Log in and manage their profile.
- Upload dental images for AI screening.
- View their AI screening results, class probabilities, confidence, and Grad-CAM visualizations.
- Browse a directory of verified doctors and book consultations.

- Attach AI reports to consultation requests and complete simulated checkouts.
- Track the lifecycle of their consultations.
- Chat with assigned doctors in real-time.
- View assessment history and download their PDF reports.
- Patients cannot access other patients' information or modify final diagnoses.

2.4 Operating Environment

Hardware Requirements:

- Processor: Intel Core i3 or above
- RAM: Minimum 8 GB
- Storage: Minimum 10–20 GB free space
- Internet: Required for web-based features and dataset access

Software Requirements:

- Operating System: Windows or Linux
- Frontend: HTML, CSS, JavaScript (React, Vite, Framer Motion)
- Primary Backend (API Layer): Node.js, Express.js
- Database: SQLite
- AI Inference Engine: Python, Flask
- Deep Learning: TensorFlow, Keras, Scikit-learn
- Image & Data Processing: OpenCV, Pillow, Pandas, NumPy
- Explainable AI: Grad-CAM
- Real-Time Messaging: Polling / Socket implementation
- Authentication: JWT

2.5 Design and Implementation Constraints

- The system shall use MongoDB as the database.
- Passwords shall never be stored as plain text.
- All protected APIs shall require authentication.
- Role-based authorization shall be implemented on both frontend and backend.

- The heavy AI inference layer must be strictly decoupled from the primary Node.js backend.
- Patients shall only access their own information.
- Doctors shall only access cases they have accepted.
- Dental images shall be validated before processing.
- Original uploaded images shall not be modified.
- Prediction results shall come from the actual trained model.
- Grad-CAM shall be generated dynamically.
- The system shall not claim medical diagnosis without verified doctor intervention.

2.6 Assumptions and Dependencies

- Required Python and Node.js dependencies are available.
- A legally usable public dental imaging dataset is available for training.
- Model performance depends on dataset quality and size.
- The system requires a trained model for actual prediction.
- If a trained model is unavailable, the application shall clearly indicate that model setup/training is required.
- Sample patient and doctor information shall be synthetic and clearly marked as demo data.

3. Specific Requirements

3.1 Data Requirements

The system shall store patient, user, doctor, consultation, assessment, report, and audit information in MongoDB.

The main data includes:

- User information
- Doctor information
- Patient information
- Dental AI assessment information
- Class probabilities
- Confidence scores
- Consultation details (status, payment status, final diagnosis)
- Grad-CAM file references
- Patient assessment history
- Reports

- Audit logs

Uploaded dental images and generated Grad-CAM images shall be stored securely with secure generated filenames. The database shall store their file references/URLs. Patient information and medical-related data shall only be accessible according to the user's role.

3.2 Functional Requirements (System Features)

The system shall provide the following main functionalities:

- **User Authentication:** Secure login and logout with role-based access for Admin, Doctor, and Patient.
- **Admin Management:** Admin can add, edit, activate, and deactivate Doctors. Admin can view system activities.
- **AI Scanning & Inference:** Patients upload dental images. The Node backend forwards this to the Python/Flask system, which validates, preprocesses, and performs AI-based classification using the trained Keras model.
- **AI Prediction & Explainability:** The system displays the predicted class, confidence score, and generates Grad-CAM visualizations to show the image regions that contributed to the AI prediction.
- **Consultation Workflow:** Patients can attach AI reports to consultation requests, process secure simulated checkouts, and doctors can manage these cases in a dedicated diagnostic workspace.
- **Clinical Verification:** Doctors can evaluate the AI's prediction, review Grad-CAM highlights, and submit an authoritative Final Diagnosis and Treatment Plan.
- **Real-Time Communication:** The system provides a real-time messaging interface between patients and assigned doctors.
- **History and Reports:** Users can view previous assessments and generate PDF reports. Patients can view only their own assessment history, Grad-CAM, and reports.
- **Audit and Error Handling:** The system records important activities and provides appropriate messages for invalid files, failed predictions, or other system errors.

3.3 Performance and ML Quality Requirements

The system shall:

- Provide a fast response for normal user operations such as login, browsing directories, and viewing records.

- Process uploaded dental images within a reasonable time (e.g., maximum response time of 2-3 seconds per image).
- Use a trained deep learning model for dental image classification.
- Display prediction confidence and uncertainty along with the AI result.
- Use the same preprocessing method during prediction as used during model training.
- Maintain reliable and consistent results for valid image inputs.
- Clearly indicate that the AI output is a screening/classification result and not a confirmed medical diagnosis.

3.4 External Interface Requirements

The system shall provide:

- **User Interface:** A responsive web interface developed using React, Vite, and Framer Motion with glassmorphism aesthetics.
- **Backend Interface:** REST APIs developed using Node.js/Express for authentication, patient management, consultations, and chat.
- **AI Interface:** REST APIs developed using Python/Flask for image processing, prediction, and Grad-CAM generation.
- **Database Interface:** MongoDB database for storing users, doctors, patients, consultations, and audit information.
- **File Interface:** Secure upload and storage of supported image formats such as JPG, JPEG, and PNG.
- **Report Interface:** PDF report generation using the assessment results, patient information, clinical diagnosis, and Grad-CAM visualization.

3.5 Security and Safety Requirements

The system shall:

- Provide secure authentication using password hashing and JWT-based authorization.
- Restrict system features according to user roles.
- Allow Patients to access only their own information and reports.
- Validate uploaded image files before processing.
- Use unique and secure filenames for uploaded files.
- Protect sensitive patient information from unauthorized access.
- Store application secrets and configuration securely using environment variables.

- Avoid exposing passwords, tokens, or unnecessary patient information in logs.
- Clearly display a medical disclaimer stating that the system is an AI-based screening tool and does not provide a confirmed medical diagnosis.

4. Appendices

4.1 Glossary of Domain-Specific Terminology

Term	Description
Dental Disease	A condition affecting the teeth, gums, or oral tissues.
Dental Image	An oral/dental photograph uploaded for AI analysis.
AI Prediction	Preliminary dental condition predicted by the trained AI model.
Confidence Score	Percentage indicating the model's confidence in its prediction.
Dental Consultation	Professional consultation between a patient and a dentist.
Deep Learning	Machine learning technique used for analyzing dental images.
Transfer Learning	Using a pre-trained model as a starting point for training on dental images.
MobileNetV2	Deep learning architecture used for dental image classification.
Classification	Process of assigning a dental image to a disease category.
Doctor Verification	Professional review of the AI-generated prediction by a dentist.
Consultation Report	Report containing the doctor's diagnosis, treatment, and follow-up details.
Grad-CAM	Technique used to visualize image regions influencing an AI prediction.

4.2 Sample Data Schema

Field	Description
Scan ID	Unique identifier for the dental scan
Patient ID	Identifier of the patient
Image	Uploaded dental image
Predicted Disease	AI-predicted condition
Confidence	Prediction confidence score
Scan Date	Date and time of analysis
Status	Current scan/consultation status

4.3 EDA Summary

The dental image dataset was analyzed to understand its structure, class distribution, missing values, and data quality. The dataset contains 12,320 dental images belonging to six disease categories: Calculus, Caries, Gingivitis, Hypodontia, Mouth Ulcer, and Tooth Discoloration. Duplicate and invalid records were removed during preprocessing before model training.